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NPS/002/005 - Technical Specification for Industrial Service Units

1. Purpose

The purpose of this document is to detail the requirements of CE Electric UK in relation to the technical requirements for Industrial Service Units for three phase systems operating at voltages up to 400V.

This document supersedes the following documents, all copies of which should be destroyed.

Ref.	Version	Title
NPS / 002 / 005	Version 1 (Nov 05)	Specification for Industrial Service Units

2. Scope

This document refers to the specification requirements of CE Electric UK with respect to Industrial Service Units (ISU) used to provide power supplies to commercial and industrial premises with loads between 100 and 500 amps per phase. The ISU shall have an integral heavy duty cut-out, pre-wired potential fuses fitted to phases and a link fitted for the neutral / earth with provision for metering CT's. They are to be suitable for internal wall mounting in public locations, but shall only allow access to internal components by appropriately authorised persons.

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3. Technical Requirements

Industrial service units (ISU's) shall meet the requirement of IEC 61439 (parts 1 and 2) and those options selected within this standard, unless specified otherwise.

3.1 Enclosure

The ISU shall be a vented enclosure and shall be finished in a subtle colour e.g. RAL 7032.

The construction of the assembly shall be such that the possibility of deliberate damage and the removal of fastenings from the outside are minimised. The assembly shall be intruder resistant, and provide protection to IP41B in accordance with IEC 60529. Ferrous metallic parts shall be protected against corrosion, sufficient to pass the tests set out in sub clause 8.2.103.1 of BS EN 60439-5 and 8.1.2 of IEC 61439-1 (severity test A).

All fastenings, nuts bolts etc. shall be plated to BS 7371 and shall have corrosion-proof locking features.

3.2 Fixing

The unit shall be suitable for wall mounting by means of a minimum of four M8 fixing bolts. All fixing holes shall be punched proud of the enclosure to allow for installation to an uneven wall surface. The upper two M8 fixing holes shall be of the 'key hole' type that readily facilitates initial positioning during installation. When installed the fixing bolts shall only be accessible from inside the enclosure.

3.3 Front Access

Access to internal components of the ISU shall be either a removable lift-off front panel or a single hinged door, which shall normally be hinged on the left-hand side and opening to a minimum of 180 degrees.

However, if the need arises at the time of installation it shall be capable of being changed to hinging on the right hand side. The panel or door shall be secured with captive screws, which shall be capable of being sealed in the closed position using standard galvanised sealing wire of maximum diameter of 2mm.

3.4 Metering cable access points

2 x pre-punched knockouts of 25mm diameter shall be provided in each side panel of the unit. The knockouts are to provide access via a flexible conduit (supplied by purchaser) between the CT coils/potential fuses and the meter operator's panel.

3.5 Cable Entry/Exit

Incoming and outgoing cables shall enter the enclosure at the bottom and top of the enclosure, respectively. To aid the installation of the incoming cable a removable gland plate complete with a cable cleat suitable for 3 or 4 core PVC covered cables ranging in size from 95mm² to 300mm² shall be provided. The cable cleat shall be so located to provide the terminating distance specified in BS 5372:1997 or greater, and shall provide adequate support to the cables both in normal service and when subject to the short circuit duty. Outgoing single core cables in the range 95mm² to 240mm², shall exit the enclosure via a hardwood or similar material top plate.

This plate shall be removable to enable fitting of cable trunking if so required.

3.6 Fuse-ways within cut out

Fixed contacts shall be accurately positioned and securely located so as to prevent misalignment during normal use. Each phase shall include a means of ensuring the fuse link is correctly aligned with the fixed contacts when closing the circuit.

Operation and alignment of the fuse links and fixed contacts shall be achieved by ensuring the fuse handle can be inserted in the distributor unit by a hinged action pivoted on the lower contact, and withdrawn by snatch action. Appropriate contact shall be made on insertion of the fuse handle so that current can be safely carried prior to tightening of the contact tightening thumbscrews. The level of protection shall meet the

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requirements of IP2X of IEC 60529 with all fuse links/handles inserted or by suitably insulated contact covers, to be provided, when withdrawn. Where fuse links/handles or contact covers are not in place, maintenance of IP XXB is not mandatory, but the design shall be such that it meets the requirements of BS EN 50274 and minimises the possibility of inadvertent contact with live conductors by the operator.

Access shall be provided for the use of test probes. They shall be able to make contact with the top and bottom fuse link terminals when fuse carriers are installed and top and bottom contacts when the fuse carrier is removed.

3.7 Fuse Carriers

Each phase of the fuse way shall be capable of independent operation. The fuse carriers shall be of a through grip shrouded type, manufactured from a suitable insulating material that provides a secure connection to the fuse link by means of wedge connections and insulated contact tightening thumbscrews. The design of the thumbscrews must ensure that when in position, the whole fuse assembly cannot be dismantled when the thumbscrews are released to their full extremities. The fuse carrier shall be suitable for accommodating fuse links with 82mm or 92mm fixing centres in accordance with BS 88 Part 5. Fuse links are **not** to be supplied with fuse carriers

The following additional tests shall be carried out on the fuse carriers:-

The test shall be carried out on each design of fuse carrier including its wedge mechanism, if any.

The test shall be made at an ambient temperature between 15°C and 25°C.

The test voltage of 3.75 kV (rms.) shall be applied for one minute between:

Both of the fuse carrier terminals without the fuse link being fitted.

One of the fuse carrier terminals and metal foil which is wrapped around surfaces of the insulating material of the fuse carrier, including thumbscrews, if any, which can be touched during live replacement or withdrawal of fuse carriers. For this test, a fuse link of the largest dimensions intended for the fuse carrier shall be fitted. Following the high voltage test, the fuse carrier shall complete a humid atmosphere test as detailed in sub-clause 8.2.4.2 of BSEN 60269 – Part 1, the insulation resistance at the conclusion of the test shall be measured; the insulation resistance measured shall not be less than 5 Megohms.

Mechanical strength - the following additional test shall be carried out - Torque tests on thumbscrews: the test shall be carried out on six random samples of the insulating thumbscrew, each complete with a wedge operating screw, to confirm the mechanical strength of the thumbscrew. Each wedge operating screw shall in turn be placed in vice and a torque applied in a clockwise direction to the insulated thumbscrew using a torque spanner adapted as necessary. Each insulated thumbscrew shall withstand a torque of 12Nm, without fracturing or turning on the shank of the wedge operating screw.

Verification of short circuit withstand strength - short time withstand tests shall be under-taken with the minimum number of distributor units to be fitted in service and, in the case of short-circuit withstand ratings verified using prospective short circuit current, the supply voltage shall be equal to 1.1 times the rated operational voltage.

3.8 Incoming Connections

The phase cores of the incoming cable shall connect to the relevant fuse way by means of range taking mechanical connectors, which are to be supplied and fitted by the supplier. The connectors shall be suitable for terminating sector shaped solid aluminium cores in the range 95mm² - 300mm² and to ensure the correct tightening torque is applied to the termination, the connector shall have range taking shear-bolts. The head of the shear-bolt shall be in the range 17mm -19mm and no specialist tooling shall be required to carry out the connection.

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3.9 Outgoing Cable Terminations

The phase cores of the outgoing cable which will normally be 95mm² - 240mm² stranded copper shall connect to the relevant outgoing fuse way by means of range taking mechanical or compression lugs supplied and fitted by the purchaser. The unit shall be supplied with suitable M12 bolts, nuts, flat and spring washers.

Suitable clearance shall be provided between phases/phase barriers to permit 2 x outgoing cores per terminal in a back to back arrangement.

The cable terminals on each outgoing fuse unit/circuit shall be separately shrouded to a minimum of IP31B. Should shrouds from one outgoing circuit be removed to gain access to its cable terminations, protection in accordance with IPXXB shall be maintained to all live parts, including bus bars and cable terminals of adjacent circuits.

3.10 Neutral / Earth

The neutral / earth bar shall be electro-tin plated brass and capable of connecting cables suitable for SNE (PE) or PME (PEN) earthing by removable links, and no earthing facility where required. Facility shall be provided for terminating sector shaped solid aluminium neutral core in the range 95mm² - 300mm², where a 4c cable is installed, and the stranded copper neutral / earth conductors of cables in this range by means of range taking mechanical connectors. To ensure the correct tightening torque is applied to the termination, the connector shall have range taking shear-bolts. The head of the shear-bolt shall be in the range 17mm - 19mm and no specialist tooling shall be required to carry out the connection.

Outgoing connection to the neutral earth shall be by means of mechanical or compression lugs supplied by the purchaser. The unit shall be supplied with suitable M12 bolts, nuts, flat and spring washers.

Facility shall also be provided for convenient connection of the CT star point to the earth bar via an M6 fixing.

3.11 Potential Fuses

Three GEC 32 or equivalent, black fuse bases and carriers without fuse links shall be mounted on a removable insulating cross member with phase identification. The bottom side shall be connected to the outgoing (upper) main fuse connections. Front top entry is required for separately supplied metering circuits. A matching white neutral link shall be provided and connected. All auxiliary wiring shall be a minimum 10mm² PVC covered conductor.

3.12 Support Bracket

A suitable bracket / shelf assembly shall be provided below the top plate to accommodate the meter operator's current transformers (CT's). This shall be suitable to accommodate 200/5, 400/5 and 500/5 CT's to BS EN 60044-1 / IEC 60044-1 and allow reasonable access for installation and termination of secondary wiring. Typically CT's are class 0.5, 7.5VA with approx dimensions of 10cm (O/D) x 6cm high.

3.13 General

3.13.1 The main current carrying components.

The main current carrying components other than those made of aluminium or aluminium alloy shall be Electro-tin or Electro-silver plated.

3.13.2 Creepage and Clearance distances.

All equipment shall be designed to ensure that adequate clearances are provided to permit a live incoming with the doors and the door(s) closed. For connections in air, the clearance and creepage distance shall not be less than 25mm between conductors and 19mm from conductor to earth.

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4. References

4.1 External Documentation

The Industrial Service Unit shall comply with the relevant International Standards, British Standard Specifications and all relevant Energy Networks Association Technical Specifications (ENATS) current at the time of tendering, except where varied by this standard. In respect the following documents are particularly relevant:

Reference	Title
IEC 61439	Low Voltage switchgear and control gear assemblies
IEC60529	Specification for the degrees of protection provided by enclosures (IP Codes)
BS 381c	Specification for colours for identification, coding and special purposes.
BS EN 60439-5	Low-voltage switchgear and control gear assemblies - Particular requirements for assemblies for power distribution in public networks
BS 7371	Coatings on Metal Fasteners
BS EN 60044-1 / IEC 60044-1	Instrument Transformers (CT's)
BS 5372	Dimensions of cable terminations for multi-core extruded solid dielectric insulated distribution cables of voltages 600/1000V and 1900/3300v having copper or aluminium conductors.
BS EN 50274	Low-voltage switchgear and control gear assemblies - Protection against electric shock - Operation by partial contact protection.
BS 88	Fuse Links
BS EN 60269	Low-voltage fuses
Part 1	General Requirements
Part 2	Supplementary requirements for fuses for use by authorised persons (fuses mainly for industrial applications)
BS 88	Cartridge fuses voltages up to and including 1000V AC. and 1500V DC.
Part 5	Specification of supplementary requirements for fuse links for use in AC. electricity supply networks

4.2 Internal documentation

Reference	Title

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4.3 Amendments from Previous Version

Clause	Subject	Amendments
Misc		References to YEDL / NEDL replaced by CE Electric UK.
2.1	Contents	Table of contents added.
3	Technical Requirements	Reference to BS EN replaced by equivalent IEC
3.1	Enclosure	BS 3382 superseded by BS 7371 (Coatings on Metal Fasteners).
3.1	Enclosure	BS EN 50300 superseded by BS EN 60439-5 (Low Voltage switchgear and Control Assemblies).
3.1	Enclosure	Reference to: Building Research Establishment Loss Prevention Standard - BRE LPS1175-5-1 level 1 deleted.
3.12	CT's	Reference made to typical CT dimensions and specifications.
3.5	Cable Entry	Reference to 4c cables added.
3.6	Fuse Ways	Reference to BS EN replaced by equivalent IEC.
3.7	Fuse Carriers	Additional testing requirements.
3.13.2	Creepage	Reference to BS 7657 and Type B fuses deleted.
4.1	External References	Updated as applicable.
Appx 1	Self Certification	Self certification tables added.

5. Definitions

Term	Definition
NIL	

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6. Authority for issue

6.1. Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
Vic Gorny	Standards Engineer	Vic Gorny	09/02/2010

6.2. Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	Head of System Strategy	M Nicholson	16/02/2010

6.3. Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
R McMahon	Standards Engineer	R McMahon	11/02/2010

6.4. DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	02/03/2010

6.5. Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	11/02/2010

6.6. Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	11/02/2010

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Appendix 1 - Supplier Requirements

Supporting evidence of compliance with type tests shall be submitted with the completed tender document.

Manufacturers may provide alternative tenders for items not complying with the above specification. This shall be clearly stated together with detailed descriptions of any variation from the specification, together with drawings and test results.

The supplier shall provide with the tender full technical details of the equipment offered and shall indicate any divergence from these standards or specifications.

Description	Catalogue Number
Industrial Service unit	044958

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Appendix 2 - Logistical requirements

To enable CE Electric UK to store the product(s) in accordance with the manufacturer's recommendations the Tenderer should provide details of the recommended storage environment with respect to each tendered product.

Details should be provided where relevant in respect to the minimum and maximum exposure levels, frequency of exposure and duration of exposure of the packaged item with respect to;

- | | | | |
|---|---------------------|---|-----------------------|
| * | Ambient temperature | * | Atmospheric corrosion |
| * | Humidity | * | Impact |
| * | Water | * | Vibration |
| * | Dust | * | Solar radiation |

The Tenderer shall ensure that each item is suitably packaged and protection to maintain the product and packaging as "fit for service" prior to installation taking account of the potential for an outdoor storage environment. All packaging shall be sufficiently durable giving regard to the function, reasonable use and contents of the packaging. Where product packages tendered are made up of sub packages all the sub packages shall unless varied by this specification, be supplied securely packaged together. Where items are provided in bagged/boxed form the material from which the bags are manufactured shall be capable of sustaining the package weight and resisting puncture by the materials within. Tenderer shall submit at the time of tendering the details of the proposed packaging (i.e. materials composition and structure) to be used for each product. Where the Tenderer is unable to provide packaging suitable for outdoor storage then this should be stated at the time of tender.

Palletised goods shall be supplied on standard 1200mm x 1000mm pallets.

Clearly legible, easily identifiable, durable and unambiguous labelling shall be applied to each individual and where relevant multiple package of like products. Where products packages tendered are made up of sub packages each sub packages shall be marked. As a minimum requirement the following shall be included;

- * Manufacturer's trademark or name
- * Supplier's trademark or name
- * Description of item
- * Date of packaging and/or batch number
- * CE Electric UK product code
- * Weight

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Appendix 3 - SELF CERTIFICATION CONFORMANCE DECLARATION

Industrial service units (ISU's) shall comply with the latest issues of the relevant national and international standards, including IEC 61439. Additionally this technical specification is intended to amplify and/or clarify requirements relating to these Standards.

This self declaration sheet identifies the clauses of the aforementioned standards relevant to Industrial Service Units (ISU's) for use on the CE Electric UK distribution network. The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non- conformance shall be entered
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

Manufacturer:

Product Reference:

**Details of the cable type(s)
suitable for use in ISU:**

Name:

Signature:

Date:

NOTE: One sheet shall be completed for each item or variant submitted.

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Technical Specification

• <u>NPS/002/005</u>			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
NPS/002/005: 3.1 Paint Finish	RAL 7032 or equivalent		
NPS/002/005: 3.2 Fixings	4 x M8 (internal) "key hole" Type		
NPS/002/005: 3.3 Front Access	Lift off front panel or single hinged 180° LHS opening door (door also suitable for RHS fixing)		
	Secured by captive screws & apposite for sealing with 2mm wire		
NPS/002/005: 3.4 Meter Cable Access	2 x 25mm pre-punched "knock outs" each side panel of unit		
NPS/002/005: 3.5 Cable Entry / Exit	Top and Bottom Entry		
	Bottom Entry suitable for 3 or 4c cables 95mm ² – 300mm ² to BS 7870 -3.40 with removable gland plate and cable cleat		

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• <u>NPS/002/005</u>			
NPS/002/005: 3.5 Cable Entry / Exit	Incoming cable terminating distance to BS 5372:1997 or larger		
	Top Entry suitable for 1c cables 95mm ² – 240mm ² via hard wood (or similar) plate (which is removable for installation of trunking if required)		
NPS/002/005: 3.6 Fuse Contacts	Shall ensure fuse link is correctly aligned with the fixed contacts		
	Insertion of fuse link by hinged pivot action on the lower contact		
	Removal of fuse link by snatch action		
	Designed so on insertion of the fuse carrier current can be safely carried prior to tightening thumbscrews		
	IP2X rating to IEC 60529 with fuse carriers inserted or if removed replaced with suitable temporary shielding		
	Provision suitable of temporary shielding		

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• <u>NPS/002/005</u>			
NPS/002/005: 3.6 Fuse Contacts	Without fuse carriers or contact covers in place preferably IPXXB to IEC 60529 and meets requirements of BS EN 50274 to minimise the possibility of contact with live conductors by the operator		
	Access for test probes to top and bottom contacts with fuse carriers in or out		
NPS/002/005: 3.7 Fuse Carriers	Supplied without fuse links		
	Capable of independent operation, "through" grip shrouded type and made from insulated material with wedge type connections and insulated tightening thumbscrews		
	Suitable for BS88 Part 5 "L" Type 82mm and 92mm centres fuse links		
	Over voltage & IR Tests		
	Mechanical Strength		
	Short circuit Withstand		

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• <u>NPS/002/005</u>			
NPS/002/005: 3.8 Incoming Connections	Shear bolt type, range taking mechanical connectors suitable for solid sector shaped aluminium conductors 95 - 300mm ² .		
	Bolt head : 17 - 19mm		
NPS/002/005: 3.9 Outgoing Connections	1 x M12 fixing for mechanical or compression connectors suitable for 1c stranded coppers conductors 95 - 240mm ²		
	Complete with fixings (nuts, bolts, plain and spring washers etc)		
	Suitable spacing for "back to back" connection of outgoing cables		
NPS/002/005: 3.9 Outgoing Connections	Shrouded to min IP31B (IPXXB if shrouding removed)		

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• <u>NPS/002/005</u>			
NPS/002/005: 3.10 Neutral / Earth	SNE (PE) or PME (PEN) earthing by removable links, and no earthing facility where required		
	As NPS/002/005: 3.8, but also suitable for stranded copper conductors in the range indicated		
	As NPS/002/005: 3.9		
	1 x M6 CT star point to the earth bar, with fixings		
NPS/002/005: 3.11 Potential Fuses/Wiring	Three GEC 32 or equiv with black fuse bases & phase identification		
	"Matching" white neutral link		
	Fuses / links mounting on removable insulated cross member		
	Outgoing connections front top entry		
	Incoming connections min 10mm ² PVC covered conductor and connected to (upper) outgoing main terminals		

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• <u>NPS/002/005</u>			
NPS/002/005: 3.11 Support Bracket / Shelf	Suitable for accommodating CT's as indicated on outgoing circuit below top plate		
NPS/002/005: 3.13.1 Main Current Carrying Components	Electro-tin or Electro-silver plated (other than those of aluminium or aluminium alloy)		
NPS/002/005: 3.13.2 Creep Electro-tin or Electro-silver plated age and Clearances	Min 25mm between conductors and 19mm from conductor to earth		

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<u>IEC 61439-1</u>			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
5.2.1 Rated Voltage of Assembly	400/230v (+10% / -6%)		
5.2.2 Rated Voltage of Circuit of Assembly	400/230v (+10% / -6%)		
5.2.3 Rated Insulation Voltage	Manufacturer to state/confirm		
5.2.4 Rated Impulse Withstand Voltage	6kV		
5.3.1 Rated Current of Assembly	500A		
5.3.2 Rated Current of Circuit Assembly	500A		
5.3.3 Rated Diversity Factor (RDF)	1.0 (3Ø outgoing circuit)		
5.3.5 Rated short-time withstand Current of Circuit of Assembly	Based on 35.5kA potential fault level and 630A BS88-2 HRC fuses at source, cable connected (300mm ² A) and 500A BS88-2 HRC fuses installed in ISU		
5.4 Rated Frequency	50Hz		
7.1.1.1 Ambient Air Temperature	≤ 40°C Av over 24hr ≤ 35°C		
7.1.2.1 Humidity	≤ 50% at 40°C		
7.1.3 Pollution Degree	2		

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• <u>IEC 61439-1</u>			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
8.1.2 Protection against Corrosion	According to 10.2.2 (severity test A)		
	8.2.103.1 of BS EN 60439-5		
8.1.3 Thermal Stability	According to 10.2.3.1		
8.1.5 Resistance to Heat and Fire	According to 8.1.5.2 And 8.1.5.3		
8.1.6 Mechanical Strength	Withstand stresses in service, during short circuit conditions and according to 10.13		
8.1.7 Lifting	According to 10.2.5		
8.2.1 Protection against Mechanical Impact	According to 10.2.6		
8.2.2 Degree of Protection	IP41B to IEC 60529		
	According to 10.3		
8.2.3 Degree of Protection of Removable Parts	Fuseways - IP2XB to IEC 60529 (if removed to BS EN 50274 & minimises inadvertent contact with live conductors by operator)		
	Outgoing Cable Ways – IP31B to IEC 60529 (IPXXB if removed)		

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• <u>IEC 61439-1</u>			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
8.3.2 Clearances	NPS/002/005: 3.13.2		
	According to 10.4		
8.3.3 Creepage	See NPS/002/005: 3.13		
	According to 10.4		
8.4.1 Protection against Electric Shock	Min IP2X to IEC 60529, otherwise IPXXB.		
8.4.3.2.2 Earth Continuity	Exposed conductive parts interconnected and connected to PE / earth conductor		
	According to 10.5.2		
8.4.3.2.3 Faults in External Circuits (Through Faults)	Conductors within assembly to withstand thermal and dynamic stresses		
	According to 10.5.3		
8.4.5 Operating and Service Conditions	Accessibility in Service by Authorised persons according to 8.4.5.2		
8.5 Switching Devices and Components	Selection, installation and accessibility according to 8.5.3, 8.5.4 and 8.5.5		
	Barriers according to 8.5.6		
	According to 10.6.1		

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• <u>IEC 61439-1</u>			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
8.6 Internal Electrical Circuits and Connections	Main Circuits in accordance with 8.6.1		
	Auxiliary Circuits in accordance with 8.6.2		
	Conductors in accordance with 8.6.3 / 8.6.4		
	Identification in accordance with 8.6.5 / 8.6.6		
	According to 10.7		
8.7 Cooling	Natural Cooling		
8.8 Terminals (Incoming and Outgoing Circuits)	According to 10.8 (see NPS/002/005: 3.8, 3.9 and 3.10)		
9.1 Dielectric Properties	Power frequency withstand in accordance with 9.1.2 and Tables 8 & 9		
	According to 10.9.2		
	Impulse withstand: 6kV		
	According to 10.9.3		
9.2 Temperature Rise	According to Table 6		
	According to 10.10		

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• <u>IEC 61439-1</u>			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
9.3 Short Circuit	Assemblies to withstand thermal and dynamic stresses resulting from short circuit currents specified		
	According to 10.11		
9.4 EMC Compatibility	According to J.9.4, Environment A (immunity & emissions for assemblies not incorporating electronic circuits)		
	According to 10.6.2		